| **Question 5** | | |
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| **(a)** | $I = \frac{E}{R} \Rightarrow \frac{\Delta I}{I} = \frac{\Delta R}{R}$ ($E$ is a constant.)<br><br>$\Delta I = 5\% \times (\frac{E}{R}) = 0.05 \times \frac{9.0\ \text{V}}{10\ \Omega} = 0.045$ A = 0.05 A (to 1 s.f.)<br><br><u>Alternative:</u><br>Max. value of $R$ = 1.05 x 10<br>Variation in $I$ = $\frac{9.0}{10} - \frac{9.0}{1.05 \times 10} = 0.0428$ A ~ 0.05 A<br><br><u>Alternative:</u><br>• min value of R - variation will be 0.0474 A, or<br>• (max – min) /2 (variation will be 0.0451 A) | **B1**<br><br>**A1** |
| **(b)(i)** | Any correct explanation, e.g.:<br>• The actual results differs from the predicted results by more than 5% (up to approximately 40%).<br>• If it was a random variation, then the actual results should deviate in such a way that it sometimes higher and sometimes lower than the predicted results (but still producing a straight line trend).<br>• Actual results is always lower than the predicted results suggesting a systematic error is present. | **A1** |
| **(b)(ii)** | As more resistance is added to the parallel network, the **effective resistance of the network of resistors reduces**.<br><br>The external circuit and the internal resistance of the battery form **a potential divider**, so a **smaller fraction of the emf is across the external circuit / the terminal p.d. decreases**, causing the measured current to be increasingly lower than the predicted current.<br><br><u>Alternative:</u><br>As more resistors are connected in parallel, **the effective resistance decreases and the current drawn increases.**<br><br>Therefore, the **potential drop across the internal resistance increases**, so the **terminal p.d. across the external circuit decreases** and the measured current becomes increasingly lower than predicted. | **B1**<br><br>**B1**<br><br>**B1**<br><br>**B1** |
| **(c)** | $E = IR + Ir \Rightarrow \frac{1}{I} = (\frac{1}{E})R + \frac{r}{E}$<br><br>$\Rightarrow \frac{r}{E} = r$(gradient) = vertical intercept<br><br>Using (1.6, 0.28) and (9.0, 1.12),<br><br>Gradient of graph = $\frac{(1.12) - (0.28)}{(9.0) - (1.6)} = 0.1135$ | **B1**<br><br>**B1** |

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| | Note to marker:<br>• Accept anything from 0.10 to 0.12.<br>• Deduct this mark if working is not shown for gradient.<br><br>$r = \frac{\text{vertical intercept}}{\text{gradient}} = \frac{(0.10)}{0.1135} = 0.88\ \Omega$<br>Accept solution if student substitutes point(s) from best-fit line and solve simultaneous equations to get the final internal resistance e.g. using two points on BFL or a point with gradient.<br><br>[-1] if the student uses data point does not lie on the BFL or use 9.0 V as the emf. | **A1** |